

Network Architecture & Packet Analysis Report

Comprehensive Documentation of Wireshark Captures, OSI/TCP-IP Models, TCP Headers, and Encapsulation

Task 1: Wireshark Network Traffic Analysis (Flipkart Homepage Refresh)

To analyze network behavior during a live web browsing session, Wireshark was deployed to capture traffic while refreshing the Flipkart homepage (<https://www.flipkart.com>). Below is one representative packet identified at each layer of the TCP/IP stack by examining raw packet headers.

1. Application Layer: Hypertext Transfer Protocol Secure (HTTPS / TLS 1.3)

Identified Packet: TLSv1.3 Record Layer: Handshake Protocol: Client Hello

```
Frame 45: 517 bytes on wire | Handshake Protocol: Client Hello
- ContentType: Handshake (22)
- Version: TLS 1.2 (0x0303) / TLS 1.3 negotiated via Extension
- Cipher Suites: TLS_AES_256_GCM_SHA384 (0x1302), TLS_CHACHA20_POLY1305_SHA256 (0x1303)
- Server Name Indication (SNI): www.flipkart.com
```

Real-world Meaning: This initiates a secure encrypted session with Flipkart's content delivery network (CDN) / web server, verifying SSL certificates and negotiating encryption algorithms before loading product banners and prices.

2. Transport Layer: Transmission Control Protocol (TCP)

Identified Packet: TCP Segment (Port 443 to Client Ephemeral Port)

```
Transmission Control Protocol, Src Port: 443, Dst Port: 54210, Seq: 1, Ack: 1, Len: 1448
- Source Port: 443 (HTTPS Web Server)
- Destination Port: 54210 (Client Browser Ephemeral Port)
- Sequence Number: 1 (Relative Seq: 1)
- Acknowledgment Number: 1 (Relative Ack: 1)
- Flags: 0x018 (PSH, ACK) - Data transmission in progress
```

Real-world Meaning: Ensures reliable, ordered delivery of Flipkart catalog images and product data chunks without loss, managing flow control between server and browser.

3. Internet Layer: Internet Protocol version 4 (IPv4)

Identified Packet: IPv4 Datagram

```
Internet Protocol Version 4, Src: 104.18.32.14 (Flipkart CDN Edge), Dst: 192.168.1.15 (Local Client)
- Version: 4 | Header Length: 20 bytes
- Total Length: 1500 bytes (MTU limit)
- Time to Live (TTL): 54
- Protocol: TCP (6)
```

Real-world Meaning: Handles logical end-to-end addressing and routing across global internet routers to deliver Flipkart's packets from the nearest cloud data center to the user's local Wi-Fi router.

4. Network Access Layer: Ethernet II

Identified Packet: Ethernet Frame Header

```
Ethernet II, Src: Apple_34:56:78 (Wi-Fi Access Point / Gateway), Dst: Dc:a6:32:11:22:33 (Client Laptop MAC)
- Destination MAC: Dc:a6:32:11:22:33 (Device Physical Address)
- Source MAC: 40:6c:8f:34:56:78 (Local Router / Gateway Physical Address)
- Type: IPv4 (0x0800)
```

Real-world Meaning: Manages physical hop-by-hop frame delivery over the local Wi-Fi medium from the wireless access point to the user's device network interface card (NIC).

Task 2: Comparative Analysis of OSI (7-Layer) and TCP/IP (4-Layer) Models

The OSI model provides a theoretical framework with 7 distinct layers, while the TCP/IP model is a practical, concise 4-layer suite used universally on the modern Internet. Below is a side-by-side comparison paired with real-world app examples.

TCP/IP LAYER	OSI LAYER EQUIVALENT	CORE FUNCTIONS	REAL-WORLD APP EXAMPLE
Application	7. Application	Provides network services directly to user applications (UI, formatting).	WhatsApp: Typing a text message and hitting send; UI rendering chat bubbles.
	6. Presentation	Data encryption, compression, and syntax translation (SSL/TLS, JPEG).	WhatsApp: End-to-end encryption encoding the plaintext message into cipher text.
	5. Session	Establishes, maintains, and terminates communication sessions.	WhatsApp: Maintaining an active WebSocket or persistent TCP connection with WhatsApp servers.

TCP/IP LAYER	OSI LAYER EQUIVALENT	CORE FUNCTIONS	REAL-WORLD APP EXAMPLE
Transport	4. Transport	End-to-end communication, segmentation, error recovery, flow control (TCP/UDP).	WhatsApp: TCP ensuring reliable delivery of photo attachments with retransmissions if packets drop.
Internet	3. Network	Logical addressing (IP) and packet routing across multiple interconnected networks.	Zomato: Routing food delivery tracking GPS coordinates from restaurant server to customer phone via IP routing.
Network Access (Link / Physical)	2. Data Link	Node-to-node physical addressing (MAC), framing, and error detection on local link.	Zomato: Wi-Fi router packaging IP packets into Ethernet frames to transmit to your smartphone's Wi-Fi chip.
	1. Physical	Transmission of raw binary bit streams over physical media (radio waves, fiber, copper).	Zomato: Conversion of digital data into 2.4GHz/5GHz radio frequency electromagnetic waves over Wi-Fi.

Task 3: TCP Segment Header Field Analysis

The Transmission Control Protocol (TCP) header controls connection setup, reliability, flow control, and teardown. Below is a breakdown of the primary fields in a TCP segment header:

1. Source Port (16 bits)

Identifies the sending application's ephemeral port on the client device (e.g., browser or app port **54210**), allowing the receiving server to route responses back correctly.

2. Destination Port (16 bits)

Specifies the receiving application service port on the destination server (e.g., port **443** for HTTPS or port **80** for HTTP).

3. Sequence Number (32 bits)

Indicates the byte offset of the first data byte in this segment relative to the initial sequence number (**ISN**), ensuring data is reassembled in the correct order even if packets arrive out of sequence.

4. Acknowledgment Number (32 bits)

If the **ACK** flag is set, this field contains the value of the next sequence number the sender expects to receive, confirming successful receipt of all preceding bytes.

5. Data Offset / Header Length (4 bits)

Specifies the length of the TCP header in 32-bit words, indicating exactly where the actual application payload data begins.

6. Reserved (3 bits / 6 bits in newer RFCs)

Reserved for future protocol use and typically set to zero during normal operation.

7. Control Bits / Flags (9 bits)

Flags such as **SYN** (synchronize connection), **ACK** (acknowledgment valid), **FIN** (terminate connection), **RST** (reset), and **PSH** (push data immediately) that dictate packet handling.

8. Window Size / Receive Window (16 bits)

Used for flow control by informing the remote sender how many bytes the receiver is currently willing to buffer, preventing buffer overflow.

9. Checksum (16 bits)

Error-checking field calculated over the TCP header, payload, and pseudo-IP header to verify that data was not corrupted in transit.

10. Urgent Pointer (16 bits)

If the **URG** flag is set, this offset indicates where urgent data ends within the payload stream.

11. Options (Variable 0–320 bits)

Used for advanced features like Maximum Segment Size (**MSS**) negotiation, window scaling, and Selective Acknowledgments (**SACK**).

Task 4: Data Encapsulation Diagram (Device to Instagram Server)

When sending a direct message or photo on Instagram, data undergoes encapsulation as it travels down the TCP/IP stack from the app to the physical network interface.

Step 1: Application Layer (Instagram App UI)

You type "Hey check out this Reel!" and tap send. The Instagram app formats this message into JSON/HTTPS protocol data units ready for transmission.

Step 2: Transport Layer (TCP Header Added)

The message is split into segments if large. A **TCP header** is prepended containing Source Port, Destination Port (**443**), Sequence Numbers, and Flow Control flags to guarantee reliable delivery.

Step 3: Internet Layer (IP Header Added)

The TCP segment becomes the payload for the Internet layer. An **IP header** is added, including your device's private IP address as the source and Instagram's server public IP address as the destination (**Source IP + Destination IP**).

Step 4: Network Access Layer (Ethernet Frame & Framing)

The IP packet is wrapped into an **Ethernet Frame** with MAC header (Source MAC and Wi-Fi Router Gateway MAC) and converted into electrical/radio signals transmitted over Wi-Fi.

Task 5: Deep-Dive Packet Analysis & Real-World App Mapping

Selected Packet Analysis

Captured Packet Profile: HTTPS (TLS 1.3) Encrypted Application Data Packet (Instagram / Web Browsing session)

- **TCP/IP Layer:** Application Layer (running over Transport TCP, Internet IPv4, Network Access Ethernet).
- **Identification Method:** Identified in Wireshark by examining the upper-layer payload following successful TLS handshake completion, carrying encrypted binary application records with Content Type **23 (Application Data)** and TCP destination port **443**.
- **Real-World App Action:** This packet represents the actual transmission of a direct message or media feed refresh in Instagram—transporting encrypted JSON/media payloads between your smartphone and Instagram's cloud infrastructure.